

Brian E. Zhang

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homepage
atomicapple0

EDUCATION

- May.'23 – May.'24 **Carnegie Mellon University** Pittsburgh, PA
Masters of Science in Computer Science, Research Thesis (GPA: 4.0/4.0)
Built LithOS, an operating system for multi-tenant deep learning workloads on GPUs.
Advised by Dimitrios Skarlatos & Todd Mowry.
- Aug.'19 – May.'23 **Carnegie Mellon University** Pittsburgh, PA
Bachelors of Science in Computer Science (GPA: 3.8/4.0; \$200,000 grant)
Coursework includes: Advanced Distributed & Operating Systems; Robot Localization & Mapping; Computer Graphics; Computer Vision; Compiler Design; HoT Compilation; ...

WORK EXPERIENCE

- Nov.'24 – Present **Modular - GPU Performance.** Software Engineer
Built core portions of MAX, a next-gen LLM Inference Engine ([Paged Attention](#) & [Prefix Caching](#), [Disaggregated Inference](#), [Speculative Decoding](#)). Also wrote some GPU kernels in [Mojo](#) and contributed to the MLIR-based AI Compiler. Much of this is [open source](#). Come take a look!
- Jun.'22 – Aug.'22 **Meta - AI Infra.** Software Engineer Intern
Improved Starlight, an internal ML pipelining platform, with compile-time type checking for Python tuple types to preemptively catch failures in user code before launching expensive training jobs. Designed customizable serdes and viz hooks for pipeline artifacts.
- Sep.'21 – Dec.'21 **NASA - Orion Backup Flight Software** Software Engineer Intern
Engineered software limits on rocket thruster firings to meet power usage requirements on the Orion spacecraft for the Artemis II mission. Built tooling to manage how bytes are packed in Orion's telemetry message structs.
- Jun.'21 – Aug.'21 **Amazon - Search Relevance** Software Developer Intern
Extended Amazon's A/B testing library to track the impact of newly released Amazon search ranking features on key business metrics. Scheduled daily Spark jobs to clean, preprocess, and extract insights from petabytes of user data.

PROJECTS

- May.'23 – Nov.'24 **LithOS: An OS for GPUs** ([Paper](#), [Slides](#)) Researcher
LithOS achieves best-in-class performance isolation and GPU utilization across many GPU sharing benchmarks. Required significant reverse engineering effort for NVIDIA GPU drivers. Written in Rust & CUDA. Work is a collaboration with Meta and is featured in [SOSP '25](#).
- Jan.'24 – May.'24 **SMoL: A SML to C Compiler** Developer
Implemented compiler passes including elaboration, hoisting, closure conversions, etc in the SML functional programming language. Built a simple runtime with garbage collector.
- Oct.'22 – Nov.'22 **Pebbles OS: A Preemptive Unix Kernel** Developer
Developed a Unix kernel from scratch in C & x86 assembly. Supports guest OSes with para-virtualization. Also wrote a POSIX-like user-space threading library on top of Pebbles.
- Mar.'22 – May.'22 **RadarSLAM: Localization for Self-Driving Cars in Adverse Weather** Developer
Wrote first open-source implementation of SOTA RadarSLAM algorithm. Evaluated algorithm performance on real-world driving datasets. [30+ GitHub stars](#).

PUBLICATIONS

Patrick H. Coppock, **Brian Zhang**, Eliot H. Solomon, Vasilis Kypriotis, Leon Yang, Bikash Sharma, Dan Schatzberg, Todd C. Mowry, Dimitrios Skarlatos. *"LithOS: An Operating System for Efficient Machine Learning on GPUs."* ACM SIGOPS 31st Symposium on Operating Systems Principles (SOSP '25). Seoul, Korea. Acceptance rate: 17.8% **IEEE Micro Top Picks Honorable Mention**: *"Most significant papers in computer architecture based on novelty and long-term impact"* for 2025.

LEADERSHIP & SERVICE

Jul.'23 – Aug.'23	Come On Out - Japan	Teacher
	Taught English to Japanese middle and high school students for five weeks in Tokyo, Nagano, and Yamanashi.	
May.'20 – May.'23	CMU School of Computer Science	Teaching Assistant
	Graded student work, wrote exams, and taught section for Principles of Imperative Computation (Summer '20) , Introduction to Robotics (Spring '23) , and Operating System Design and Implementation (Fall '23) . Received overwhelmingly positive student feedback. Read reviews here .	
Dec.'21 – May.'23	CMU Explorer's Club	Quartermaster
	Maintained club's outdoor equipment and hosted weekly gear checkouts for members.	
Dec.'20 – Jan.'22	CMU Puzzlehunt	Staff & Puzzle Writer
	Organized and wrote the biannual CMU Puzzlehunt for over 1500 participants. My puzzles include: Mother Functions , The Pirate's Gambit , A Tartan's Responsibility .	
Aug.'20 – May.'23	CMU Recreational Running Club	Treasurer
Dec.'20 – May.'21	CMU Housing Services	Resident Assistant

AWARDS

* = team competition

2025	Honorable Mention	IEEE Micro Top Picks
2024	3rd Place	CMU Algorithms With A Purpose AI Contest*
	\$250 Recipient	CMU Robotics Club SHRG Grant
2023	University Honors	CMU
2022	1st Place (2-way tie)	CMU Robot Arm Autonomous Jenga Contest*
	1st Place, \$1000 Prize	CMU Mobile Robots Race (Video)
2020	Category Prize	CMU TartanHacks*
2019	"Ring of Honor"	CMU Intro Comp. Biology, Research Project
	10th Place	FAMAT Programming Contest*
2018	\$2000 Recipient	Mu Alpha Theta Grant
	Alumni	Wolfram Summer School
2017	2nd Place	NSU Psychology Bowl*

LANGUAGES

fluent English
conversational Mandarin

PROGRAMMING

C, Rust, Python, Java,
CUDA, MLIR/LLVM, SML,
Mojo, Why3, MATLAB,
Mathematica, Scala,
Docker, Bash, Git, $\LaTeX$

INTERESTS

puzzlehunts, 2d
animation, biking, shogi,
cooking, board games,
pickleball